

# Kimiya Jozani

Email: kimiya@vt.edu  
LinkedIn: kimiya-jozani  
Website: kimiyajozani.github.io  
Phone: +1 202-830-5891

## Education

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- ✧ **PhD in Industrial and Systems Engineering**, Virginia Tech *2023-Present*
- ✧ **MEng in Industrial and Systems Engineering**, Virginia Tech *2023-2024*
- ✧ **B.Sc. in Industrial Engineering**, Sharif University of Technology *2016-2021*

## Selected Graduate Coursework

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- ✧ Stochastic and Robust Optimization, Integer Programming, Linear Optimization, Non-linear Optimization, Random Processes, Simulation, Machine Learning, Management of Organizational Systems, Socio-technical Systems

## Research Interests

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- ✧ **Methodology:** Stochastic Optimization, CVaR Optimization, Machine Learning
- ✧ **Application:** Epidemic Management, Invasive Species Control, Healthcare Policy, Behavioral Economics

## Publications

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- ✧ Escobar, L., Jozani, K., Romero-Alvarez, D., Brennan, R.N., Ulloa-Zepeda, L., Vial, C., Buyuktahtakin, E. Epidemiology of Andes virus (Orthohantavirus andesense). Under Review at The Lancet.
- ✧ Jozani, K., Sageer, N. A., Eldardiry, H., Tunc, S., & Buyuktahtakin, E. (2025). A Multi-echelon Demand-driven Supply Chain Model for Proactive Optimal Control of Epidemics: Insights from a COVID-19 Study. IISE Transactions. (Preprint available on ArXiv).
- ✧ Choi, S. J., Jozani, K., Cooper, J. F., & Buyuktahtakin, E. (2025). Learning to Optimize at Scale: A Benders Decomposition-TransfORMers Framework for Stochastic Combinatorial Optimization. NeurIPS 2025 Workshop on Machine Learning and Operations Research (MLxOR).
- ✧ Galande, N., Jozani, K., & Buyuktahtakin, E. (2025). Artificial Intelligence in Supply Chain Optimization: A Systematic Review of Machine Learning Models, Methods, and Applications. Under review at Annals of Operations Research.
- ✧ Yan, K., Jozani, K., & Buyuktahtakin, E. (2025). AI-Driven Spatio-Temporal Forecasting and Management of Wayleaf Grass. Workshop paper Presented at the 20th INFORMS Workshop on Data Mining and Decision Analytics (DMDA).

## Ongoing Research

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- ✧ Hantavirus Modeling, Joint work with Escobar, L., and Buyuktahtakin, E. (May 2026 - Present)
- ✧ Psychiatric setting operations simulation and optimal control, Joint work with Carilion Clinic (February 2026 - Present)
- ✧ Learning to decompose optimization problems. Joint work with Fioretto, F., and Buyuktahtakin, E. (August 2025 - Present).
- ✧ Sample complexity in CVaR problems. Joint work with Khodadadian, S., and Buyuktahtakin, E. (April 2025 - Present).
- ✧ Epidemic management under distributional uncertainty. Joint work with Tunc, S., and Buyuktahtakin, E. (March 2025 - Present).

## Peer Review

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- ✧ Invited Review for IISE Transactions journal *July 2026*
- ✧ Invited Review, Naval Research Logistics journal *April 2026*

## Teaching Experience

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- ✧ Instructor of record for Probability Foundations for ISEs, Virginia Tech, US *2025*
- ✧ Graduate Teaching Assistant, Virginia Tech, US *2023 – 2025*
  - Global Issues in Industrial Management (ISE 4304) *Spring 2023*
  - Organizational Processes and Design (ISE 4004) *Fall 2023*
  - Data Management for Industrial and Systems Engineers (ISE 2034) *Spring 2024*
  - Probabilistic Operations Research (ISE 3414) *Fall 2024*
  - Optimization: Linear and Nonlinear Programming (ISE 5405) *Fall 2024*
  - Industrial Automation (ISE 4264) *Spring 2025*
- ✧ High School Mathematics Teacher, Pajooresh High School, Iran *2016- 2022*

## Research Experience

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- ✧ Systems Optimization and Machine Learning Lab (SysOptiMaL), Virginia Tech *Dec 2023 - Present*
  - Epidemic management: Large-scale, multi-echelon supply chain optimization *Dec 2023 - Aug 2025*
  - Epidemic management: Prediction and control of Avian Influenza outbreaks in US farms *Feb 2026 - Present*
  - AI-driven optimization: Developing a learning-to-decompose framework *Aug 2025 - Present*
  - AI-driven optimization: Learning-enhanced optimization algorithm design *May 2024 - Present*
  - Invasive species management: Optimal control of Wavyleaf in Virginia *Aug 2025 - Present*
  - Healthcare management: Assessment of no-shows and cancellations in psychiatric settings *Feb 2026 - Present*
- ✧ Human Technology Interaction Research, Virginia Tech *Mar 2022 - Dec 2023*
- ✧ Consumer Behavior Analysis During COVID-19, Sharif University of Technology *Jun 2020 - Feb 2021*

## Academic Presentations

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- ✧ Jozani, K., Tunc, S., Buyuktahtakin, E. “A Decision-Dependent Distributionally Robust Optimization Model for Proactive Mitigation of Infectious Disease Outbreaks.” INFORMS 2026 Annual Meeting Job Market Showcase Track. November 2026.
- ✧ Jozani, K., Tunc, S., Buyuktahtakin, E. “A Decision-Dependent Distributionally Robust Optimization Model for Proactive Mitigation of Infectious Disease Outbreaks.” MSOM Conference. July 2026.
- ✧ Jozani, K., Khodadadian, S., Buyuktahtakin, E. “A General Framework for Finite-Sample Evaluation of CVaR Optimization.” INFORMS 2026 Optimization Society Conference. March 2026.
- ✧ Jozani, K., Choi, S. J., Cooper, J. F. Buyuktahtakin, E. “Learning to Optimize at Scale: A Decomposition-TransfORMers Framework for Stochastic Combinatorial Optimization.” 2025 NeurIPS MLxOR Workshop. December 2025.
- ✧ Jozani, K., Tunc, S., Buyuktahtakin, E. “A Multi-Stage Dynamic Optimization Framework for Effective Epidemic Resource Allocation and Sustained Public Trust.” INFORMS 2025 Annual Meeting. October 2025.

- ✧ Jozani, K. “Proactive and Need-based Epidemic Resource Allocation Through an Integrated Supply Chain Framework: Insights from a COVID-19 Study.” Invited Presentation at the David Eccles School of Business at The University of Utah. September 2025.
- ✧ Jozani, K., Buyuktahtakin, E. “Equitable Vaccine Allocation through a Dynamic Epidemic-Supply Chain Model.” INFORMS MIF 2025 Summer Webinar Competition. July 2025.
- ✧ Jozani, K., Buyuktahtakin, E. “Equitable Vaccine Allocation through a Dynamic Epidemic-Supply Chain Model.” VT GPSS Research Symposium. March 2025.
- ✧ Jozani, K., Buyuktahtakin, E. “Equitable Vaccine Allocation through a Dynamic Epidemic-Supply Chain Model.” VT Montgomery Distinguished Lecture Poster Presentation. March 2025.
- ✧ Jozani, K., Buyuktahtakin, E. “Equitable Vaccine Allocation through a Dynamic Epidemic-Supply Chain Model.” INFORMS 2024 Annual Meeting Health Applications Society Invited Presentation. October 2024.
- ✧ Jozani, K., Buyuktahtakin, E. “Equitable Vaccine Allocation through a Dynamic Epidemic-Supply Chain Model.” 2024 CeZAP Infectious Diseases Research Symposium. October 2024.

## Grants

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- ✧ Track 2 Seed Grant from Whole Health Consortium at Virginia Tech (Student PI) 2026
- ✧ Invasive Species Collaborative (ISC) at Virginia Tech Graduate Student Grant Program (co-PI) 2025-2026
- ✧ VT Center for Emerging, Zoonotic, and Arthropod-borne Pathogens ID IGEP Graduate Student Grant (PI) 2025

## Awards

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- ✧ VT-CMB Travel Award 2025
- ✧ Agee GTA Award for the best graduate teaching assistant 2024
- ✧ Best Poster Award at VT Grado ISE Graduate Poster Competition 2024
- ✧ ISE Grado Admission Travel Award 2022

## Honors

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- ✧ Selected for the IISE Future Faculty Fellows (3F) Program 2025
- ✧ Nominated and acknowledged as an outstanding undergraduate mentor at Virginia Tech 2024
- ✧ Ranked in the top 10% of Industrial Engineering graduates, Sharif University 2021
- ✧ Ranked in the top 1% in the Iran National University Entrance Exam 2016

## Software and Programming

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- ✧ Languages (Python, C++); Solvers/Modeling (Gurobi, CPLEX, AMPL); Statistics (Minitab, SPSS); Simulation (AnyLogic); Other (Tableau, Visual Paradigm, SolidWorks)

## Academic Services

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| ✧ VT INFORMS President                                              | <i>2026 - 2027</i> |
| ✧ VT ISC Graduate Student Grant Subcommittee Member                 | <i>2026</i>        |
| ✧ VT INFORMS Vice President                                         | <i>2025 - 2026</i> |
| ✧ Virginia Tech ISE Senior Symposium Poster Judge                   | <i>2024 - 2025</i> |
| ✧ Operations Research Liaison at Women in ISE (WISE), Virginia Tech | <i>2024 - 2026</i> |
| ✧ INCOSE Board Member at Virginia Tech                              | <i>2023 - 2025</i> |

## Student Mentorship

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| ✧ Kaiyu Yan, ISE PhD Student at Virginia Tech                   | <i>2025-Present</i> |
| ✧ Seung-Jin Choi, ISE PhD Student at Virginia Tech              | <i>2025-Present</i> |
| ✧ Matthew Horne, ISE Undergraduate Student at Virginia Tech     | <i>2026-Present</i> |
| ✧ Matthew Asbury, ISE Undergraduate Student at Virginia Tech    | <i>2026-Present</i> |
| ✧ Joon Yong Shim, ISE Undergraduate Student at Virginia Tech    | <i>2026-Present</i> |
| ✧ Nihal Sageer, ISE Accelerated Master Student at Virginia Tech | <i>2024-2026</i>    |
| ✧ Joshua Cooper, ISE Undergraduate Student at Virginia Tech     | <i>2025</i>         |

## Professional Experience

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| ✧ IT Manager, Pajooresh High School, Iran          | <i>2019-2022</i> |
| ✧ Project Manager, Karena Innovation Factory, Iran | <i>2020</i>      |